

Telegram Workflow — Test Guide

For Zenith Education Studio · Two-person test: Admin (Ajitesh) + Parent (Friend) · Generated 2026-05-10

SECTION 1 — BEFORE YOU START (ADMIN)

Complete all items below before calling your friend. If any are missing, fix them first.

- ☐ **ngrok is running** — terminal shows `Forwarding https://unshackle-idealism-cardstock.ngrok-free.dev → http://localhost:3000`. If it's not, run `ngrok http 3000` in a new terminal.
- ☐ **Dev server is running** — `npm run dev` is active in a separate terminal. Visit `http://localhost:3000` to confirm.
- ☐ **NEXT_PUBLIC_APP_URL is set** — `.env.local` has `NEXT_PUBLIC_APP_URL=https://unshackle-idealism-cardstock.ngrok-free.dev`. If you restarted the dev server since last session, verify this is still there.
- ☐ **Telegram bot is connected** — open `http://localhost:3000/dashboard/settings` and confirm the Telegram card shows "Connected".
- ☐ **Admin Telegram is linked** — Settings → Admin notifications card shows "Linked". If not, generate a fresh linking code and send `/linkadmin <code>` to the bot.
- ☐ **Google Calendar is connected** — Settings → Google Calendar shows "Connected". Required for Tests 3 and 4.
- ☐ **3 test classes exist** — open `http://localhost:3000/dashboard/classes` and confirm: *P5 Mathematics @ Tampines Hub (seats available)*, *Sec 3 A-Math @ Bishan Junction 8 (FULL)*, *JC1 H2 Mathematics @ Jurong East JEM (seats available)*.
- ☐ **Send your friend the bot link** — share `https://t.me/<YourBotUsername>` with your friend and ask her to open it and tap Start. She should see a greeting (or blank; doesn't matter). Have her ready to chat.

SECTION 2 — TEST CASES

TEST 1

Basic FAQ — Bot answers a general question

Verifies the core RAG + chat engine works over Telegram.

PARENT (YOUR FRIEND)

1. Open Telegram, find the bot.
2. Send: *Hi, what subjects do you offer?*
3. Send: *How much is P5 Math?*

ADMIN (YOU)

1. Watch — no action needed for this test.
2. Confirm bot responds within ~5 seconds.
3. Check `/dashboard/conversations` — a new Telegram conversation should appear.

✓ **Pass if:** Bot replies with subject list and mentions P5 Math at \$320/month, Wed 5–6:30pm, Tampines Hub. Conversation appears in dashboard with channel badge "Telegram".

TEST 2

Scope Guard — Bot refuses off-topic questions

Verifies the bot stays in lane and doesn't hallucinate answers about unrelated topics.

PARENT (YOUR FRIEND)

1. Send: *What's the weather like today?*
2. Send: *Can you help me with my child's homework?*
3. Send: *What do you think of other tuition centres?*

ADMIN (YOU)

1. Watch the bot replies.
2. Confirm it does NOT attempt to answer any of these.

✓ **Pass if:** Bot politely redirects all three with something like "Sorry, I can only help with things related to Zenith Education Studio — fees, classes, bookings and so on."

TEST 3

Trial Booking (Happy Path) — Parent books P5 Math

Full booking flow: bot collects details, confirms, writes booking row, creates Google Calendar event, sends admin ping.

⚠ **Before starting:** Tell your friend to give realistic-ish info but use your own real email as the parent email so you can verify the calendar invite arrives.

PARENT (YOUR FRIEND)

1. Send: `Hi, I'd like to book a trial for P5 Math`
2. Bot will ask for details one at a time. Provide:
 - Child's name: **Sarah**
 - Your name: **Mrs Tan**
 - Phone: **+6591234567**
 - Email: *(use your real email so you get the invite)*
3. When bot summarises and asks "Shall I lock that in?" — send: `Yes please`
4. Note the confirmation message — it should include date, time, and Tampines Hub address.

ADMIN (YOU)

1. Watch your Telegram for a 🎉 **New trial booking** ping. Should arrive within seconds of your friend confirming.
2. Check the ping contains: Sarah, P5 Mathematics, Tampines Hub, next Wednesday 5–6:30pm.
3. Open `/dashboard/bookings` — new row should appear with "Cal. synced" badge and centre "Tampines Hub".
4. Check your Google Calendar — event "Trial: Sarah — Mathematics (P5)" on next Wednesday 5pm SGT.
5. Check your email — Google Calendar invite should have arrived from the calendar.

✓ **Pass if:** (1) Admin gets Telegram 🎉 ping. (2) Booking row in dashboard with synced badge. (3) Calendar event on correct date/time with Tampines Hub as venue. (4) Parent's Telegram receives the calendar invite line "I've also sent a calendar invite to...".

TEST 4

Waitlist — Parent tries to book the full Sec 3 A-Math class

Verifies the bot detects a full class and routes the parent to the waitlist instead of booking.

PARENT (YOUR FRIEND)

1. Send: `I want to book Sec 3 A-Math for my son`
2. Provide details when asked (son's name: **Ryan**, phone: **+6598765432**).
3. When bot offers the waitlist, say: `Yes, please add us to the waitlist`

ADMIN (YOU)

1. Watch for 📌 **New waitlist entry** ping in your Telegram. Check it shows Ryan + position number.
2. Open `/dashboard/waitlist` — Ryan should appear under the Sec 3 A-Math group with his waitlist position.

✓ **Pass if:** Bot does NOT create a booking (it's full). Instead routes to waitlist. Admin gets 📌 ping. Waitlist row appears in dashboard with correct position number.

TEST 5

Escalation + Admin Takeover — Parent complains, admin steps in

Full escalation loop: parent is upset → bot flags it → admin gets pinged → admin takes over → admin replies directly to parent via dashboard → parent receives it in Telegram → admin resolves.

⚠ **Before starting:** Make sure you're logged into the dashboard at `http://localhost:3000/dashboard/conversations` and can see it during the test.

PARENT (YOUR FRIEND)

1. Send: `I'm very unhappy — the teacher was late last week and nobody apologised`
2. Bot should reply with something like "I understand your concern, let me connect you with our team..."
3. Wait for a direct message from the admin (this comes via the same bot chat).
4. Reply to whatever the admin sends.

ADMIN (YOU)

1. Watch your Telegram — a 🚨 **Parent escalation** ping should arrive with the parent's message and a link to the conversation.
2. Tap the link — opens `/dashboard/conversations/<id>`.
3. Click **"Take over"** — amber banner appears: "You've taken over this conversation".
4. Type in the reply box: `Hi Mrs Tan, this is the centre manager. I'm so sorry to hear this — let me look into it right away. Can I call you back in the next 30 minutes?`
5. Click **Send**.
6. Watch for your friend's reply to appear *live* in the conversation viewer (without refreshing).
7. Once resolved, click **"Mark resolved"**.

✓ **Pass if:** (1) Admin gets 🚨 ping. (2) Conversation viewer shows "Take over" button. (3) After takeover, bot stops auto-replying. (4) Admin's typed reply appears in your friend's Telegram chat (sent by the bot but with your words). (5) Friend's next message appears in the dashboard viewer in real time (Supabase Realtime). (6) After "Mark resolved", conversation status in the list changes to "closed".

Note on real-time: The friend's reply should appear in the dashboard without you refreshing the page. If it doesn't appear, wait 5 seconds — Supabase Realtime can have a short connection warmup.

TEST 6

PDPA Opt-Out — Parent sends STOP

Verifies the opt-out flow closes the conversation and stops the bot from responding.

PARENT (YOUR FRIEND)

1. Send: `STOP`
2. Try sending another message after: `Actually wait, I have a question`

ADMIN (YOU)

1. Check `/dashboard/conversations` — the conversation should be marked "closed".
2. After the second message, the bot should reply with "This chat has been closed..." and not engage further.

✓ **Pass if:** STOP gets confirmation reply. Conversation marked closed in dashboard. Follow-up message gets a static "closed" response, not a bot answer.

TEST 7

Web Widget → Telegram Opt-In Deep Link

Verifies that a parent who books on the website can tap a link to connect their Telegram for future reminders.

⚠ **You do this one alone.** Your friend doesn't need to be involved.

YOU — ACTING AS PARENT

1. Open `http://localhost:3000/test.html` in a browser.
2. Start a fresh conversation (click ↺ reset button).
3. Book a JC1 H2 Math trial: provide name, phone, and a real email address.
4. Confirm the booking.
5. Look for the footer block in the bot's confirmation reply:
 *Get reminders & updates on Telegram*
→ `https://t.me/YourBot?start=b_<uuid>`
6. On your phone, tap that link → Telegram opens → tap Start.
7. Bot should reply: "✅ You're linked. I'll send reminders and updates here for Sarah's trial."

YOU — VERIFYING AS ADMIN

1. Check `/dashboard/bookings` — new booking row should appear.
2. In Supabase Studio, open **students** table → find the student just created → check `telegram_chat_id` column is now populated.

✓ **Pass if:** Web booking confirmation shows the Telegram deep-link line. Tapping it opens the bot. Bot confirms the link. Student row in DB has `telegram_chat_id` set.

SECTION 3 — RESULTS TRACKER

#	Test	Result	Notes
1	Basic FAQ	TO TEST	
2	Scope guard	TO TEST	
3	Trial booking (happy path)	TO TEST	
4	Waitlist flow	TO TEST	
5	Escalation + admin takeover	TO TEST	
6	PDPA opt-out	TO TEST	
7	Web widget → Telegram deep link	TO TEST	

SECTION 4 — IF SOMETHING GOES WRONG

Symptom	Likely cause	Fix
Friend messages bot, no reply	ngrok tunnel expired or dev server crashed	Check ngrok terminal — restart if needed. Check dev server terminal. Then re-send the message.
Bot replies but takes 30+ seconds	OpenAI API cold start or slow connection	Normal on first message. Should be ~3–5s after warm-up.
🎉 admin ping doesn't arrive	Admin Telegram not linked or bot token expired	Settings → Admin notifications → re-link via <code>/linkadmin</code> .
Calendar event not created	Google Calendar disconnected or token expired	Settings → Google Calendar → disconnect and reconnect.
Admin reply doesn't reach friend	Telegram bot token or channel_user_id mismatch	Check dev server terminal for <code>[telegram webhook] send failed</code> . Reconnect the bot in Settings.
Realtime doesn't update dashboard	Migration 009 not applied or Realtime not enabled	Run migration 009 in Supabase. Then Supabase Studio → Database → Replication → confirm <code>messages</code> and <code>conversations</code> are in the publication.
Deep link from web doesn't show bot	<code>telegram_bot_username</code> not populated	Settings → Telegram → disconnect → reconnect. This backfills the username.

SECTION 5 — QUICK REFERENCE

URLs you'll need

- Dashboard: `http://localhost:3000/dashboard`
- Conversations: `http://localhost:3000/dashboard/conversations`
- Bookings: `http://localhost:3000/dashboard/bookings`
- Waitlist: `http://localhost:3000/dashboard/waitlist`
- Settings: `http://localhost:3000/dashboard/settings`

Test classes (Zenith)

- **P5 Mathematics** — Wed 5–6:30pm, Ms Tan, Tampines Hub, \$320/mo, seats available
- **Sec 3 A-Math** — Sat 10–11:30am, Mr Lim, Bishan Junction 8, \$380/mo, **FULL**
- **JC1 H2 Mathematics** — Tue 7–9pm, Dr Wong, Jurong East JEM, \$480/mo, seats available

- **Widget test:** `http://localhost:3000/test.html`

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